Clustering Identification and Error Propagation for Oort Constants and Milky Way Galactic Curve Derived From GAIA DR3

1. Introduction

The study of galactic dynamics has led to the scientific acceptance of dark matter by Vera Rubin in the 1970s, with her 1970 paper on galactic rotation indicating another source of mass and her definitive survey of the phenomenon in 1980 solidifying the idea. Put simply galactic dynamics is the study of how a galaxy moves internally. This field utilizes stellar dynamics to predict and fit internal movement of a galaxy (Binney and Tremaine, 2008). As the search to better understand dark matter, galactic dynamics is one of our best tools, it can explain how galaxies evolve (UW Madison, 2026) and even help test the laws of physics at large scales (Milgrom, 1983).

Rotation curves of a galaxy are a mapping of the radial velocity of stars at varying radii from a galactic center (Hedemaki, 2017). The first Milky Way rotation curve was mapped by Horace Babcock in 1940 (Babcock, 1940). **Talk about advancements here and recent papers.** The rotation curves are useful because they allow astrophysicists to fit their models of how a specific galaxy should move to real data (Binney and Tremaine, 2008). For example: **Model Derived from paper using jeans here. (Source and Figure)**.

While a rotation curve can help describe a galaxy's general rotation. A set of four constants can describe in more detail the movement of a specific portion of stars in the galaxy. Regarding the Sun's local area these are known as the Oort constants. These constants describe the first-order Taylor expansion of a two-dimensional velocity field. They describe the azimuthal shear, vorticity, radial shear, and divergence of stars in a local area.

An important bias in galactic dynamics is the presence of galactic clusters and streams. These are defined as a group of stars that share an origin and are gravitationally bound to a meaningful degree. These are important as they can bias the data by acting as outliers (Bloch et al, 2025). Unless specifically looking into their dynamics, they serve only to shift the data. This is further enunciated in the fact that the certainty of data biases us to select a limited sample of data, making these outliers all the more swaying in the Oort constant and Galactic Curve derivation.

Astronomical data is cluttered with measurement uncertainty. Thus the refined data such as Oort Constants and the Galactic Curve also maintain degrees of uncertainty. Often magnified by the process. Precise understanding of the Oort Constants and the Galactic curve is important because they define the stellar kinematics of the milky way at a local level and a global level respectively. (**Motivation here).** Large clusters and streams present the opportunity for further systematic biases. Therefore it is important to look at identifying these systematic biases and seeing how they affect the refined data (Andreon and Hurn, 2012). While also keeping into account the model uncertainty with which we identify the biases.

Astrophysics data has already nearly outgrown the realm of manual processing (Garafolo et al. 2017). Furthermore it is outgrowing the capability for supervised learning of machine learning models (Fotopoulo, 2024).

This paper will generally cover three types of error. These are measurement uncertainty, which derive from the certainty of the equipment used to gather the data in the GAIA DR3, systematic biases, which arrive from the aforementioned clusters and streams, and model uncertainty, which is the certainty to which our methods can calculate the Oort constants and the rotation curve….**Challenges+Gaps/Proposed Solutions**

Thus we look to unsupervised learning as a means to improve certainty among astrophysics calculations by evaluating how stellar clusters and tidal streams serve to bias the data and further bias the processed data. In this paper we will address the use of cluster identification and removal methods in calculating the Oort Constants and the Milky Way's Rotation Curve, both used to describe stellar kinematics at a local and galactic level respectively (Hedemaki, 2017).

We take a separate approach to derive the Oort constants from the galactic curve. For the galactic curve we use a simple regression on the equation derived using jeans equations (source here). However for the local Oort Constants we will use a maximum likelihood estimation with a log likelihood function. To identify the maximum we will use a Markov chain Monte Carlo method. To identify the stellar clusters and stream we will use a spectral clustering algorithm

2. Data and Methodology

2.1 GAIA DR3 Data

GAIA DR3 was collected by the ESA space observatory currently in orbit at the L2 point. This data takes the form of a table of points of interest in the sky. These are stars, quasars, galaxies, etc. These celestial objects have an uncertainty given along with them as well as the full 5-parameter astrometric solution for 1.46 billion objects. (Gaia, 2022**)**

For the two different galactic features, we collect and filter the data in two distinct ways. For the galactic curve we go about the approach taken by **Source here** and use an elongated ellipse from the galactic center, through the sun solar system, and towards the outer galactic disc. We then use this ellipse to only search through stars in this area and with a parallax uncertainty below **Insert final plax uncertainty bound here**. This is due to its inherent covariance with other component uncertainty such as proper motion and radial velocity (El-Badry, 2025).

However for the local Oort constants we take a different approach for selecting data to stay consistent with its definition (Binney and Tremaine, 2008). We select stars within a radius of **Final Number here** from the sun, however we use the same parallax uncertainty bounds as with the galactic curve data

After taking the GAIA DR3 data from each star, in equatorial phase space, we use the transformation equations and constants defined by the International Celestial Reference System (IAU, 2000) to transform to galactic phase space.

2.2 Spectral Clustering

After filtering and transforming the data to a usable state, we use a spectral clustering method to identify strong stellar clusters. We take the approach utilized by (Almiedo and Prieto, 2013). First the Laplacian of the data graph is calculated. The distance metric is defined solely via kinematics with the 6D phase space vector. With the similarity function being fit within a range of 0-1 with the gaussian.

Figure 1

$$S(a,b) = e^{-d(a,b)^2}$$

Where d(a,b) is the euclidean distance metric in 6D phase space

After calculating and using the eigenvectors of the Laplacian, the groups are recalculated using a simple K-means algorithm. Utilizing the elbow method we can find the optimal k value where we are not overfitting to the data but we also find the correct amount of clusters. This method compares the Within-Cluster Sum of Squares (WCSS), which simply calculates the total squared distance between a mean and its datapoints, value and number of clusters by finding the sharp bend in slope. We also seek to avoid reaching a local minimum, which can be common for these simple machine learning algorithms (Nadler and Galun, 2006). So we attempt the clustering multiple times from random starting positions. We can now use the WCSS metric to determine the compactness of the various groups found algorithmically (Cui, 2020). We will take the approach of utilizing different levels of cleaned data sets. The proceeding calculations will be completed independently on each data set where the varying factor is the filtering of stellar clusters. The threshold WCSS metric to be retained in the data set will be varied from trial to trial.

2.3 Rotation Curve Analysis

After creating the various cleaned data sets we proceed to calculate the local Oort constants and the galaxy rotation curve. For the rotation curve we will simply recover a plot of circular velocities and attempt a simple fitting per data set. The fitting occurs on a curve derived using simple high school physics. First gravitational force is equated to centripetal force and the complicated part comes in when modeling the mass distributions within a radius. The model is typically split into a luminous matter and dark matter distributions. The dark matter profile is often modeled with the NFW model, derived by Navarro in 1996, (Navarro et al., 1996).

Figure 2

The luminous matter is then further split into to pieces, the mass contributed from the spherical galactic bulge at the center of a galaxy and the thin stellar disk. (**Source)**

Figure 3

We will fit the data sets to find the parameters of these models using a simple regression.

2.4 Oort Constant Estimation

The local Oort constants will be more involved. For the calculated local Oort constants we will proceed using the same process improved upon by Jo Bovy of the University of Toronto with the GAIA DR1(Bovy, 2017). First there is a need to look at the proper motion equations, which are functions of three observable data inputs and seven parameters.

Figure 4

$$\mu_l(l, b, \varpi) = (A\cos 2l - C\sin 2l + B)\cos b + \varpi(u_0 \sin l - v_0 \cos l)$$

$$\mu_b(l, b, \varpi) = -(A\sin 2l + C\cos 2l + K)\sin b\cos b + \varpi[(u_0\cos l + v_0 \sin l)\sin b - w_0 \cos b]$$

Of these seven, three are known qualities of our solar system, (u_0,v_0,w_0). They represent the velocity of our sun towards the galactic center, the velocity towards galactic rotation, and the velocity towards the galactic north pole. The remaining four are the Oort constants A, B, C, and K. We can then create the log likelihood function modified by Bovy (Bovy, 2017)

Figure 5

$$\ln \mathcal{L} = -\frac{1}{2}\sum_i \left(\frac{[\mu_l^i - \mu_l(l_i, b_i, \varpi_i]^2}{\sigma_{\mu_l}^2 + (s_{\mu_l}^i)^2} + \ln[\sigma_{\mu_l}^2 + (s_{\mu_l}^i)^2] + \frac{[\mu_b^i - \mu_b(l_i, b_i, \varpi_i]^2}{\sigma_{\mu_b}^2 + (s_{\mu_b}^i)^2} + \ln[\sigma_{\mu_b}^2 + (s_{\mu_b}^i)^2]\right)$$

We can then use Markov chain Monte Carlo for finding the likely parameters in phase space. These parameters will then be the local Oort constants we are searching for.

3. Results

[Cluster identification data here]

[Rotation curve data]

[Oort constant data]

4. Discussion

5. Conclusions / Future Directions

Citations

Add here from noodletools once paper finished